



JPK

**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

**KOD DAN NAMA PUSAT
BERTAULIAH**

SOALAN PENILAIAN PENGETAHUAN (SUBJEKTIF)

KOD UNIT CORE ABILITY

NAMA UNIT ABILITY

NAMA CALON

NO.KAD PENGENALAN

MASA

TARIKH

MARKAH

Arahan kepada calon:

1. Tulis nama dan nombor kad pengenalan calon pada ruangan yang disediakan;
2. Calon tidak dibenarkan membuka kertas soalan sehingga dibenarkan;
3. Calon hendaklah menjawab semua soalan;
4. Calon dilarang membawa nota atau sebarang bahan rujukan kecuali yang dibenarkan.
5. Dilarang meniru semasa penilaian; dan
6. Dilarang membawa keluar kertas soalan dari bilik peperiksaan

KERTAS PENILAIAN INI MENGANDUNGI02.... MUKA SURAT BERCETAK

QUESTION 1

1. Every workplace are required to have a safety and health plan and program to prevent accidents or unwanted incidents from occurring. The management must establish a clear goal and objective for the safety and health programme. This is to ensure that established safety and health programme can be continuously evaluated. The goal and objective has to be communicated clearly to all employees – failure of which will result in less employee participation in the programme.

- (a) List at least three(3) benefits of HSE awareness programmes (6 marks)
- (b) Give at least two(2) ways that management can provide visibility and involvement in implementing the safety and health (4 marks)
- (c) Briefly explain three (3) ways to control the exposure of chemical to the workers at the workplace and methods can be applied. (6 marks)
- (d) Name at least four(4) types of environmental problems that mainly can be contributed by the industries? (4 marks)

QUESTION 2

You are engaged as a safety and health officer in a factory which produce chemical base product. The first task to be conducted is to plan and implement HSE awareness programme in the factory.

- (a) Explain four(4) main points to consider when planning health, safety & environment awareness programme. (4 marks).
- (b) Explain four(4) program and at least three(3) of its contents which can be proposed for the success of health awareness program. (10 marks)
- (c) What is programme evaluation and the benefit of evaluating the HSE awareness programme? (6 marks)

SCHEMA

Question 1

a) It enable workers to:

- recognize unsafe practices and conditions. It requires following of safety, health and environment procedures.
- get to know all about the conditions that promote health, safety and environment at the workplace.
- be willingness to take time to prevent accident.
- have the right attitude about health, safety and environment, both for their own and their coworker's.
- prevent injury, illness and accidental damage to company property.
- increased workers morale, satisfaction and motivation
- Improved alertness and concentration among workers

b) Management can provide visibility and involvement in implementing the safety and health through;

- Commitment to safety and health policy
- Establishment of safety and health consultative committee
- involvement in providing resources for safety and health programme

c) Three (3) ways to control the exposure of chemical to the workers at the workplace and methods can be applied. are;

(i) Control at Source– Method; substitution, Modify process, Enclosure, Local exhaust, Fugitive, emission control, Isolation

(ii) Control at Path– Methods; Guards, Isolation, Enclosure & Engineering Control, Housekeeping, Continuous area, monitoring, automation or remote control

(iii) Control at receiver– Methods; by Law & Regulations, Administration and PPE

d) Types of environmental problems that mainly can be contributed by the industries are;

- a) Global warming
- b) Air pollutions
- c) Water pollutions
- d) Noise pollutions
- e) Soil pollutions

Question 2

a) The main points to consider are when planning HSE awareness program:

i) Support and resources

The overall success of the initiative depends on the cooperation, acceptance and involvement of everyone in the organizations. Early and ongoing support from senior management is key to the success of the programme. Support from top and senior management is critical and should be visible to all staff in the organisation. When management leads the program and requests support from workers, the credibility of the program is enhanced. Communication from senior management promoting the program is one way of demonstrating commitment.

ii) Know current situation

To select or decide any HSE awareness programme to be conducted, designated team or personnel must be assessed:

- current HSE situation at the workplace in terms of compliance to law and procedures; accident, dangerous occurrences and health illness statistics;
- current awareness level among workers regarding issues related to HSE at workplace; and
- categories of workers which the HSE issues have been determined and identified as a problem

iii) Target audience profile and goals

It is important to confirm who the target audience is for intended program because message and delivery mechanisms will differ according to intended audience. It is equally important to recognize and consider the varying circumstances, needs and concerns that each group may have if you will be targeting different groups. Clearly defining audiences will allow development of effective messages, activities and plans that will address the needs of each group.

iv) Communication plan

Communication is vital in ensuring successful of intended awareness programme. There are many resources that can be considered to support and enhance communication efforts. Right language for the target audience must be used. Use of technical language should be avoided, unless addressing a technical audience. Keep it simple and appropriate, easy to understand and remember, consistent and relevant, focused on one idea and repeated frequently. Consider using a workers briefing or meeting as the forum for the first communication.

b) Four(4) program which can be proposed for the success of health awareness program are;

(i) Chemical control program

Employer has to develop and carry out programmes spelling out methods and procedures on safety and health in the use of chemicals at workplace which include;

- a. Cooperation employer and employee at the workplace
- b. Establish Inventory of chemicals - Chemical Register
- c. Conduct Chemical Health Risk Assessment(CHRA)
- d. Ensure Purchasing procedure, Classification and Labelling
- e. Ensure Safety Data Sheet for all chemical hazardous to health
- f. Ensure the adequacy of control measures is implemented

ii) Conduct Hearing Conservation Program (HCP)

HCP is a set of carefully planned activities carried out with the aim of protecting and maintaining

the hearing of employees. Aspect of HCP is based on requirements under the Noise Exposure Regulations 1989

- a) Conduct noise monitoring
- b) Control of noise or use of ppe
- c) Established audiometry test
- d) Training and education program
- e) Record keeping

iii. Ergonomics Programme

The purpose of the Ergonomics Program is to establish a system for the identification, evaluation, and control of ergonomic risk factors in the workplace. (1)

Components of Ergonomic Programme include;

- c) Responsibilities of Management/Employer Employees - Workers Supervisors, Occupational Health Services, Environment, Health, and Safety Committee (EHS)
- d) Program Element - Worksite Analysis and Prevention and Intervention Strategies
- e) Record Keeping - for evidence of injuries or disorders associated with cumulative trauma disorders.

iv. Confined Space Program

Working in a confined space can expose a person to a very risky situation that can cause diseases and injury which leads to an accident and death.

The confined space entry programme shall include at least the following elements

- a) **Hazard identification** - Identify all hazards associated with working in the confined space.
- b) **Risk assessment** - Carry out the risk assessment before carrying out work involving entry into a confined space

- c) **Risk control** - Based on the risk assessment, establish and implement the procedures and practices of hazard control that provide safe entry into the confined space.
- d) **Stand-by person** - At least one stand-by person outside the confined space into which entry is authorised for the duration of entry operations.
- e) **Monitoring of multiple confined spaces** - For monitoring of multiple confined spaces by a single stand-by person, the programme shall include the means and procedures to enable the stand-by person to respond to an emergency affecting one or more of the confined spaces being monitored without distraction from the stand-by person's responsibilities
- f) **Permit system** - Develop a written system for preparation, issuance, implementation, and cancellation of permit to work.
- g) **Sign-posting** - Confined space warning signages to be posted at conspicuous places when work is being carried out after the issuance of the permit to work.
- h) **Employee training** - Ensure that employees who are involved in confined space operation have undergone training on confined space safety course approved by the Director General of DOSH and passed the examination.
- i) **Equipment** - Provide appropriate equipment use for testing and monitoring; personal protective emergency and rescue equipment; and any other equipment necessary for safe entry into and rescue from confined spaces
- j) **Means of Communication** – such as telephone or walkie talkie
- k) **External hazard protection** - Protection such as physical barriers to be provided to control potential hazards posed by pedestrians and vehicles.
- l) **Identification of the duties** - identify the duties of employees to be an authorised entrant, stand-by person, entry supervisor, authorised gas tester, permit issuer or other related personnel in entry operation.
- m) **Information to the contractor**

(c) **Awareness program evaluation** is a systematic method for collecting, analyzing, and using information to answer questions about programs, particularly about their effectiveness and efficiency. In both the public and private sectors, stakeholders often want to know whether the programs they are funding, implementing, voting for, receiving or objecting to are producing the intended effect. While program evaluation first focuses around this definition, important considerations often include how much the program costs per participant, how the program could be improved, whether the program is worthwhile, whether there are better alternatives, if there are unintended outcomes, and whether the program goals are appropriate and useful. In other words, evaluation of awareness programmes is:

- to gain information on how to improve future awareness programs.
- to justify the existence and budget of the training department by showing how it contributes to the organization's objectives and goals; and
- to decide whether to continue or discontinue training programs.